Kanato MATSUTANI

Curriculum Vitae

Doctoral student, Quark Physics Laboratory, Physics Program, Division of Advanced Science and Engineering, Graduate School of Advanced Science and Engineering, Hiroshima University
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Science and Technology Specialties

- **Field:** Experimental nuclear and hadron physics; detector research and development for the Electron-Ion Collider (EIC) / ePIC experiment.
- **Silicon timing detectors:** AC-coupled Low Gain Avalanche Detectors (AC-LGAD), strip and pixel types, for the ePIC Barrel Time-of-Flight (BTOF) and Zero Degree Calorimeter; timing resolution (tens of picoseconds) and position resolution characterization; sensor quality assurance (QA).
- **Laser test systems:** design and commissioning of a Transient Current Technique (TCT) system (1064 nm pulsed IR laser, motorized XY stages, high-bandwidth digital oscilloscope DAQ, multi-channel strip readout).
- **Beam tests:** electron beam tests at KEK (PF-AR), including temperature-dependence studies of timing response; data acquisition operation.
- **Electronics:** front-end readout ASIC development (at BNL) and readout board development for strip-type AC-LGADs.
- **Data analysis & simulation:** Python-based waveform analysis pipelines, digital signal processing, timing-resolution budget decomposition, position reconstruction; detector simulation studies for the ePIC BTOF.
- **Sensor inspection:** optical microscopy (KEYENCE VHX) and automated edge-chipping detection for semiconductor wafer QA.

Education and Professional History (chronological, no gaps)

Apr 2017 – Mar 2020	Student, Aichi Institute of Technology Meiden High School, Nagoya, Aichi, Japan. High school diploma, March 2020.
Apr 2020 – Mar 2024	Undergraduate student, Department of Physics, School of Science, Hiroshima University, Higashi-Hiroshima, Hiroshima, Japan. **Bachelor of Science (Physics)**, March 2024.
Apr 2024 – Mar 2026	Master's course student, Physics Program, Division of Advanced Science and Engineering, Graduate School of Advanced Science and Engineering, Hiroshima University, Higashi-Hiroshima, Hiroshima, Japan. **Master of Science**, March 2026.
Apr 2024 – present	Researcher (graduate student appointment), International Institute for Sustainability with Knotted Chiral Meta Matter (WPI-SKCM^2), Hiroshima University, Higashi-Hiroshima, Hiroshima, Japan (concurrent with graduate study).
Nov 2024 – Feb 2025	Visiting researcher, Brookhaven National Laboratory, Upton, NY, USA, under the Ozaki Exchange Program (KEK US-Japan Science and Technology Cooperation Program). Front-end ASIC development for AC-LGAD readout (11 Nov 2024 – 8 Feb 2025; concurrently enrolled in master's course).
Apr 2026 – present	Doctoral course student, Physics Program, Division of Advanced Science and Engineering, Graduate School of Advanced Science and Engineering, Hiroshima University, Higashi-Hiroshima, Hiroshima, Japan (Quark Physics Laboratory; supervisor: Prof. Kenta Shigaki).
Apr 2026 – present	Next-Generation Fellow, Program for Developing and Supporting the Next-Generation of Innovative Researchers at Hiroshima University (JST SPRING); appointment through Mar 2029.
Jan 2027 – Mar 2027 (planned)	Visiting researcher, Physics Division, Brookhaven National Laboratory (ePIC), under the Ozaki Exchange Program 2026.

Theses

Mar 2026 M.Sc. thesis: “Performance Evaluation System of AC-coupled Low Gain Avalanche Detector (AC-LGAD) for Precise 4D Tracking at the EIC-ePIC Experiment,” Hiroshima University.

Mar 2024 B.Sc. thesis: “Design of a performance evaluation system for new LGAD detectors and performance evaluation of the readout circuit for high spatiotemporal resolution measurements in the EIC ePIC experiment,” Hiroshima University (written in Japanese; title translated).

Publications

- K. Matsutani and K. Shigaki for the ePIC-Japan, “Performance evaluation system for AC-coupled low gain avalanche detector (AC-LGAD) for precise 4-dimensional tracking in EIC-ePIC experiment,” *RIKEN Accelerator Progress Report 2025* (to be submitted).

Conference Presentations

International

Nov 2026 (accepted) “From AC-LGAD Time Resolution to PID Reach at the EIC: Test-Beam and TCT Laser Studies for the ePIC Barrel Time-of-Flight,” oral, Asian Triangle Heavy-Ion Conference (ATHIC 2026), Shanghai, China.

Jul 2026 “AC-LGAD sensor QA in Japan: new infrastructure & first laser results,” oral, July 2026 Joint ePIC/EICUG Meeting, University of Glasgow, UK.

Apr 2023 “Simulation studies and R&D status and plans in Japan for the EIC ePIC Barrel TOF with AC-LGAD,” poster, The 9th Asian Triangle Heavy-Ion Conference (ATHIC 2023), Hiroshima, Japan. K. Matsutani, W. Yamauchi, Y. Akiba, Y. Goto, T. Hachiya, I. Nakagawa, K. Shigaki, S. Yano for the EIC Japan group.

Domestic (talks at JPS meetings given in Japanese; titles translated)

Sep 2026 “Temperature dependence of the timing response of AC-LGADs for the EIC-ePIC BTOF detector evaluated with an electron beam,” oral, JPS 81st Annual Meeting, The University of Tokyo (Komaba). K. Matsutani *et al.* for ePIC-Japan.

Mar 2026 “Basic characterization of strip-type AC-LGADs for the EIC-ePIC experiment,” oral, JPS 2026 Spring Meeting (online). K. Matsutani, M. Kaneta, T. Gunji, K. Shigaki, T. Hachiya, S. Yano for ePIC-Japan.

Sep 2025 “Evaluation of AC-LGAD Time Resolution for Precision Particle Identification in the EIC-ePIC Experiment,” oral, HEHI-ex-J Workshop, Kumamoto.

Sep 2025 “Evaluation of AC-LGAD time resolution for high spatiotemporal resolution measurements with particle identification detectors in the EIC-ePIC experiment,” oral, JPS 80th Annual Meeting, Hiroshima University.

Mar 2025 “Development of an AC-LGAD system for high spatiotemporal resolution measurements with particle identification detectors in the EIC ePIC experiment,” oral, JPS 2025 Spring Meeting (online).

Sep 2024 “Performance evaluation of AC-LGAD detectors for high spatiotemporal resolution measurements in the EIC ePIC experiment,” oral, JPS 79th Annual Meeting, Hokkaido University.

Aug 2024 Same title as above, poster, Tutorial Workshop on the Physics of High-Energy Heavy-Ion Collisions, Osaka University.

Mar 2024 “Performance evaluation methods and prospects of AC-LGAD detectors for high spatiotemporal resolution measurements at the EIC,” poster, JPS 2024 Spring Meeting (online). K. Matsutani, K. Shigaki, S. Yano, P. Tribedy, A. Tricoli.

Sep 2023 “Performance evaluation method of new LGAD detector and prospects at Hiroshima University,” oral, ALICE/sPHENIX/STAR/EIC-J Workshop, RIKEN.

Awards, Fellowships and Grants

Oct 2026	Ozaki Exchange Program 2026 (KEK), selected.
Apr 2026	Next-Generation Fellowship, JST SPRING at Hiroshima University (Apr 2026 – Mar 2029).
Mar 2026	Excellent Master's Thesis Presentation Award, Physics Program, Hiroshima University.
Jul 2025	Tuition exemption for outstanding master's students (FY2025), Hiroshima University.
Mar 2025	Doctoral Course Advancement Scholarship; commended as an “Especially Outstanding Student,” Hiroshima University.
Sep 2024	Ozaki Exchange Program 2024 (KEK), selected; research stay at BNL, Nov 2024 – Feb 2025.

Research Visits, Beam Tests and Schools

Jul 2026	Hamamatsu Photonics K.K. (Hamamatsu): AC-LGAD performance report and next-production meeting.
May – Jun 2026	KEK beam test: pixel-type AC-LGAD for ePIC-ZDC; temperature dependence of strip-type AC-LGAD for ePIC-TOF.
Mar 2026	YITP International School on EIC Physics, Yukawa Institute for Theoretical Physics, Kyoto University.
Feb 2026	KEK beam test: DAQ commissioning; pixel/strip AC-LGAD and AstroPix data taking.
Jul 2025	EICUG & ePIC Joint Collaboration Meeting and Early Career Workshop, Jefferson Lab, USA.
Jun – Jul 2025	2nd European School on the Physics of the EIC, Benicàssim, Spain.
Jul – Aug 2024	Particle Physics Computing Summer School (PPCC-SS-2024), KEK; WPI-SKCM² Summer School (Hokkaido Univ./Hiroshima Univ.).
Jan 2024	EIC-Asia Workshop, National Cheng Kung University, Taiwan.
Dec 2022, Aug 2023	KEK: training on AC-LGAD test-bench setup and performance evaluation methods.

Collaboration Membership

- ePIC Collaboration (Electron-Ion Collider); ePIC-Japan group.

Last updated: October 2026